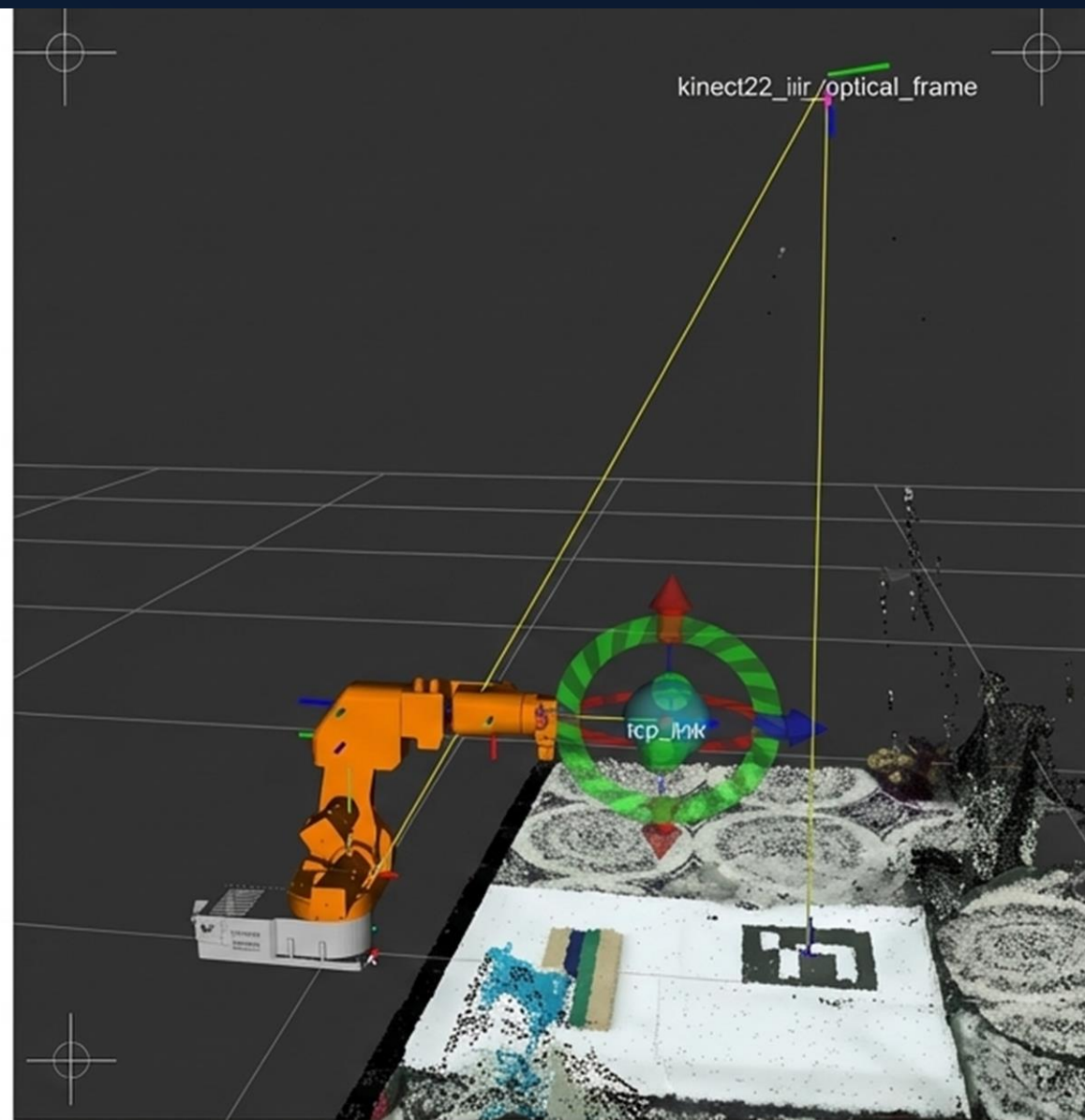


Vision System

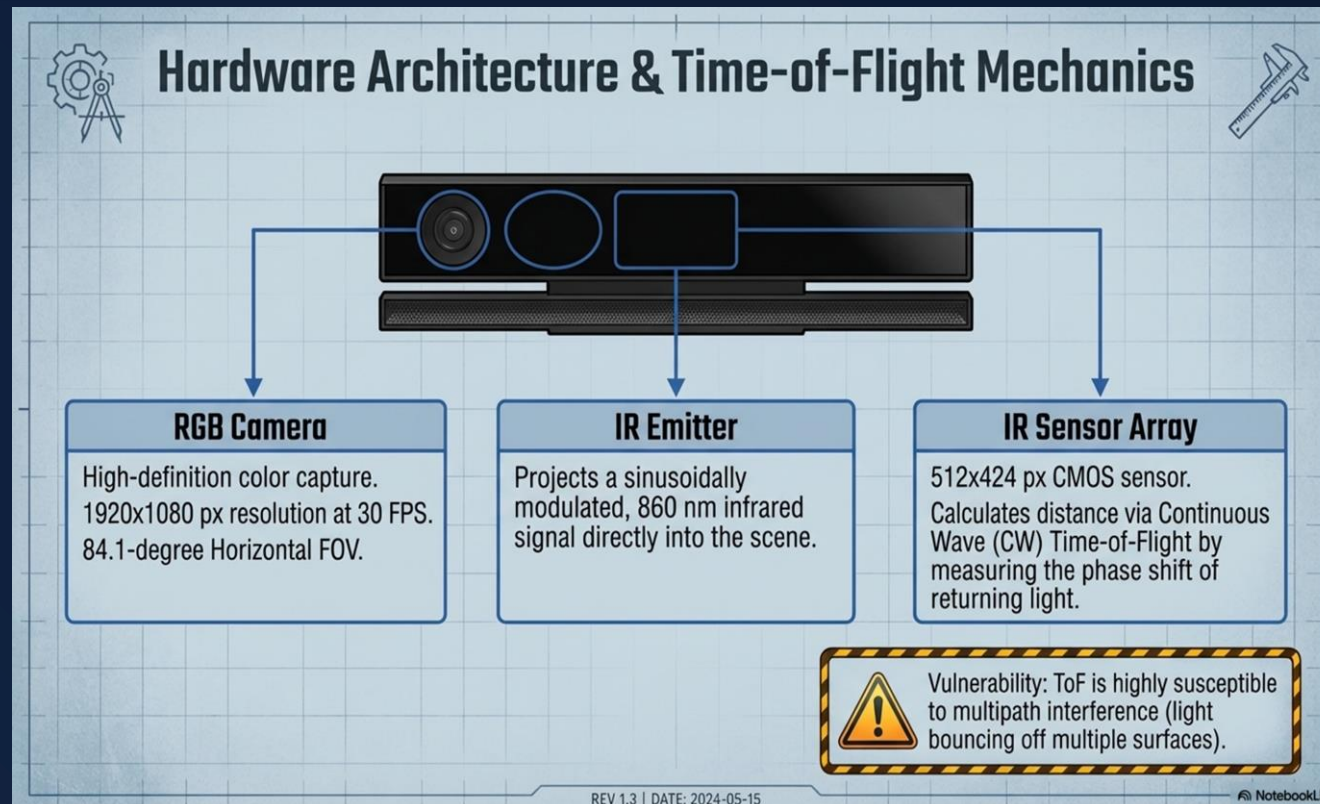
Integrating & Calibrating the Kinect v2

From Raw Time-of-Flight Data to Sub-Millimeter 3D Trajectory Generation



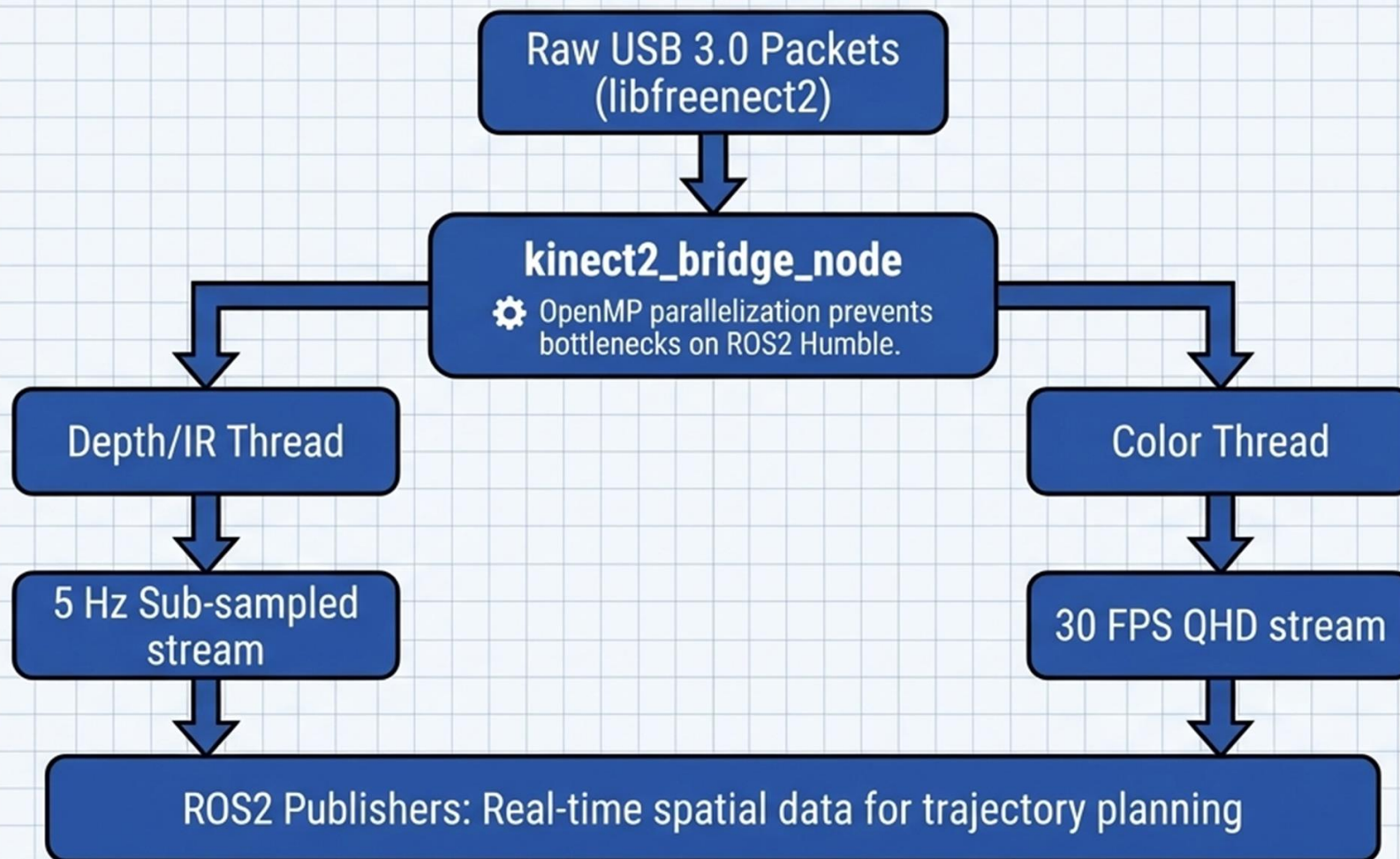
KINECT V2 HARDWARE & ARCHITECTURE

The **Microsoft Kinect v2**, originally developed for the Xbox One entertainment system, has become a cornerstone in the field of low-cost 3D sensing and robotics vision. Unlike its predecessor, which utilized structured light technology, the Kinect v2 employs the Continuous Wave (CW) Time-of-Flight (ToF) principle to generate high-resolution depth maps [1]. This transition to **ToF** significantly improves depth accuracy, particularly in environments with varying lighting conditions, making it an ideal candidate for precision tasks such as continuous-path tracking and automated welding detection.



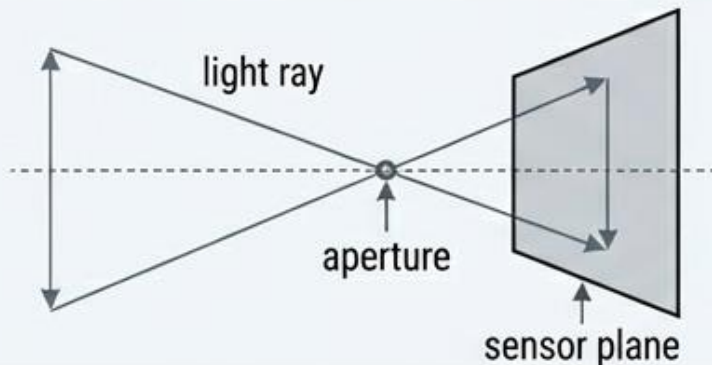


The Middleware: Parallelizing Data via kinect2-ros2



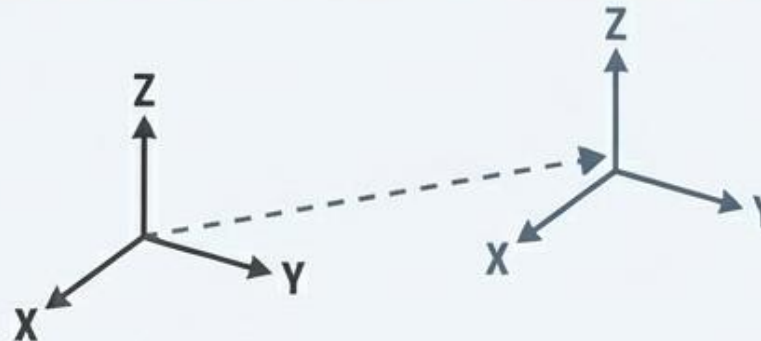
The Anatomy of Calibration: Intrinsics vs. Extrinsics

Intrinsic Calibration



- **Target:** Individual camera lenses.
- **Goal:** Modeling the ideal pinhole camera to mathematically eliminate lens warpage.
- **Outputs:**
 - Focal lengths (f_x , f_y)
 - Principal points (c_x , c_y)
 - Radial distortion (k_1 , k_2)
 - Tangential distortion (p_1 , p_2)

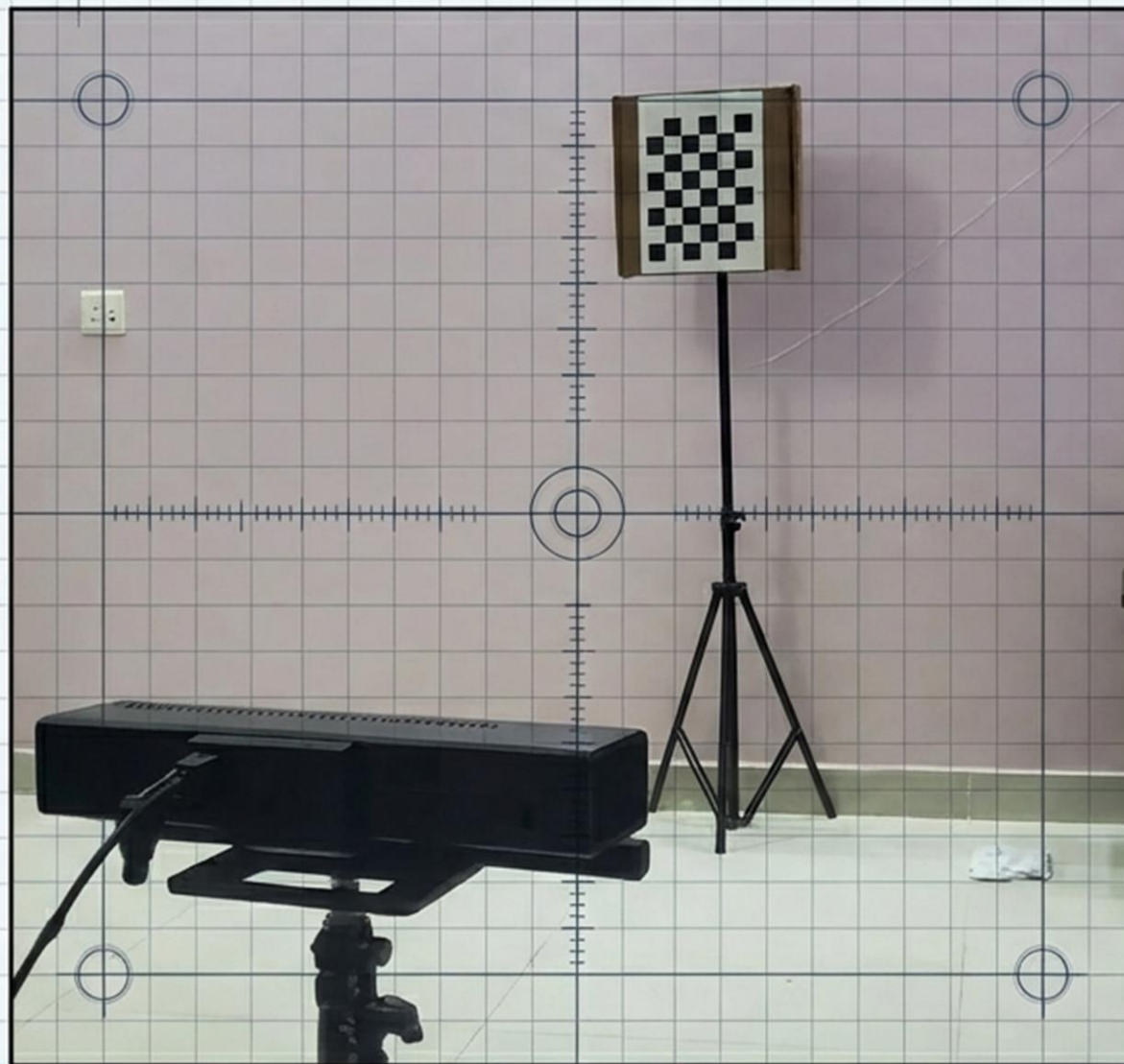
Extrinsic Calibration



- **Target:** Relationship between RGB and IR sensors.
- **Goal:** Spatial 6-DOF alignment to generate perfectly registered RGB-D point clouds.
- **Outputs:**
 - Translation Vector ($\mathbf{T}$)
 - Rotation Matrix ($\mathbf{R}$)

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Stage 1: Hardware Setup & Environment



PRE-FLIGHT CHECKLIST

1. Absolute Rigidity

The 5x7x0.03m chessboard pattern must be hard-mounted to a flat surface. Any physical warping ruins the mathematical solve.

2. Controlled Positioning

Secure the sensor on a locked tripod at a precise distance of 1.0 to 1.5 meters from the target.

3. Diffuse Lighting

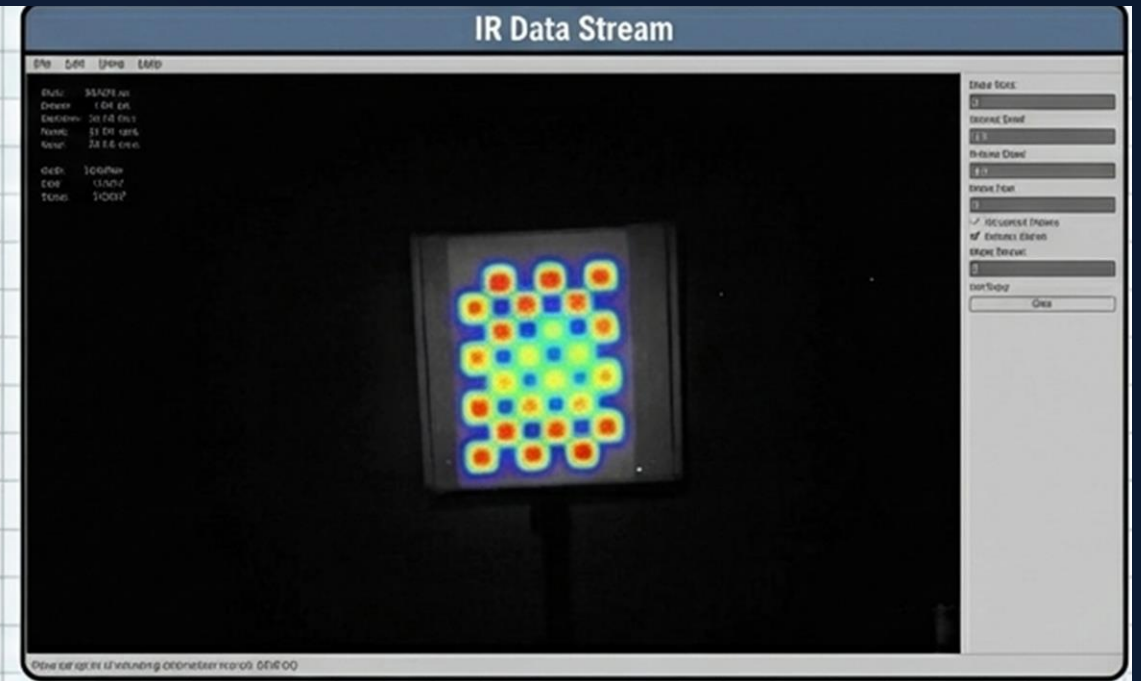
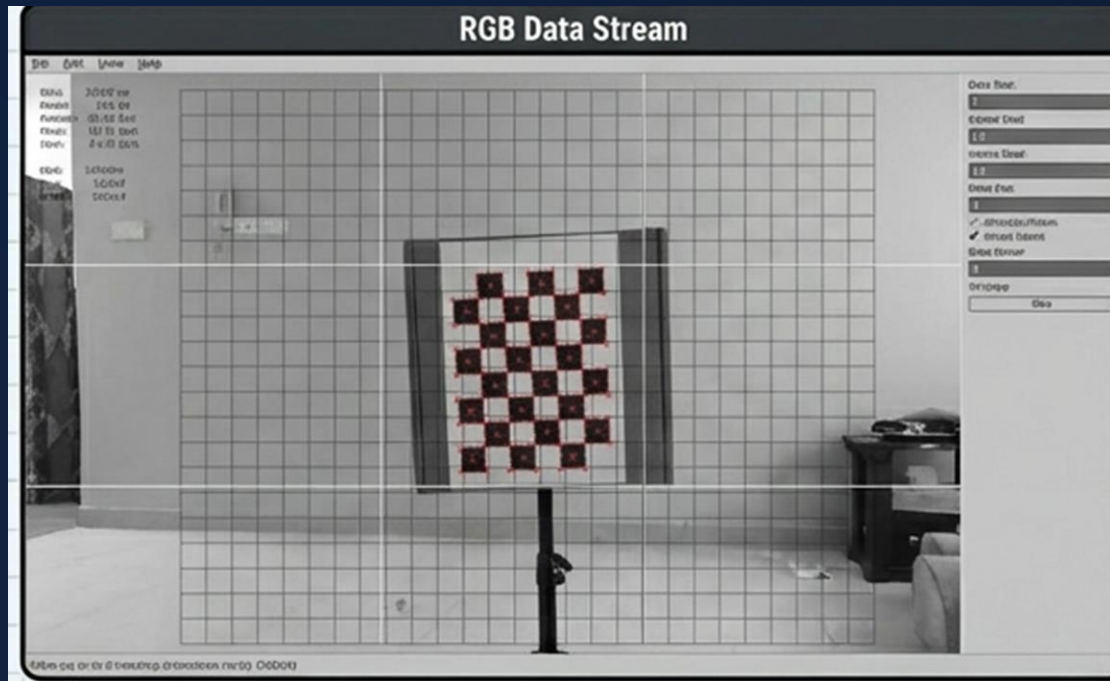
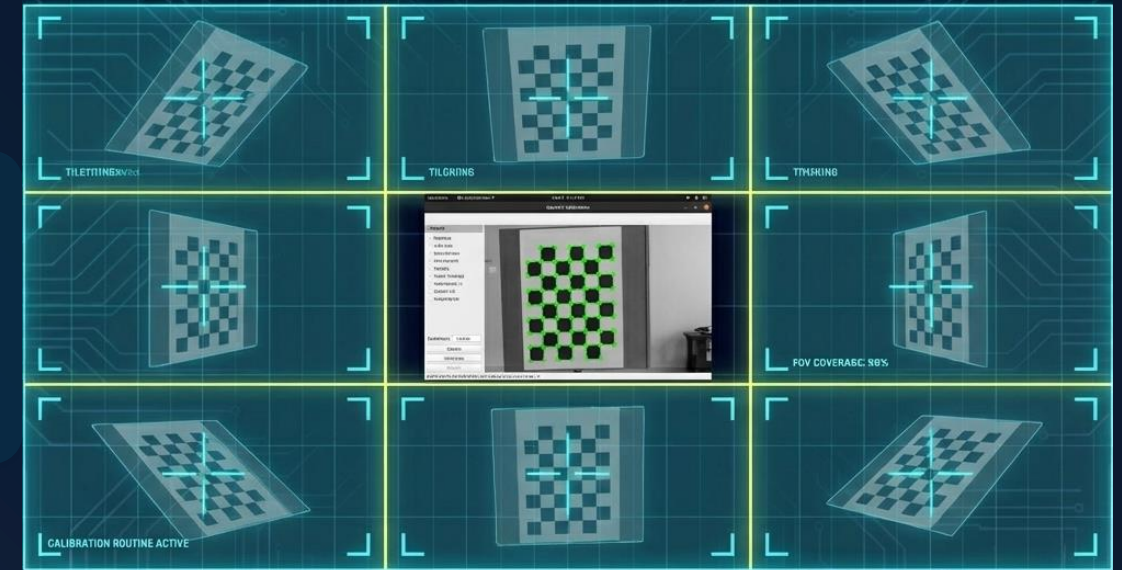
Employ even, consistent lighting to prevent IR glare or saturation on the black squares, ensuring the grid is simultaneously visible to both RGB and IR lenses.

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Stage 2: Data Collection:

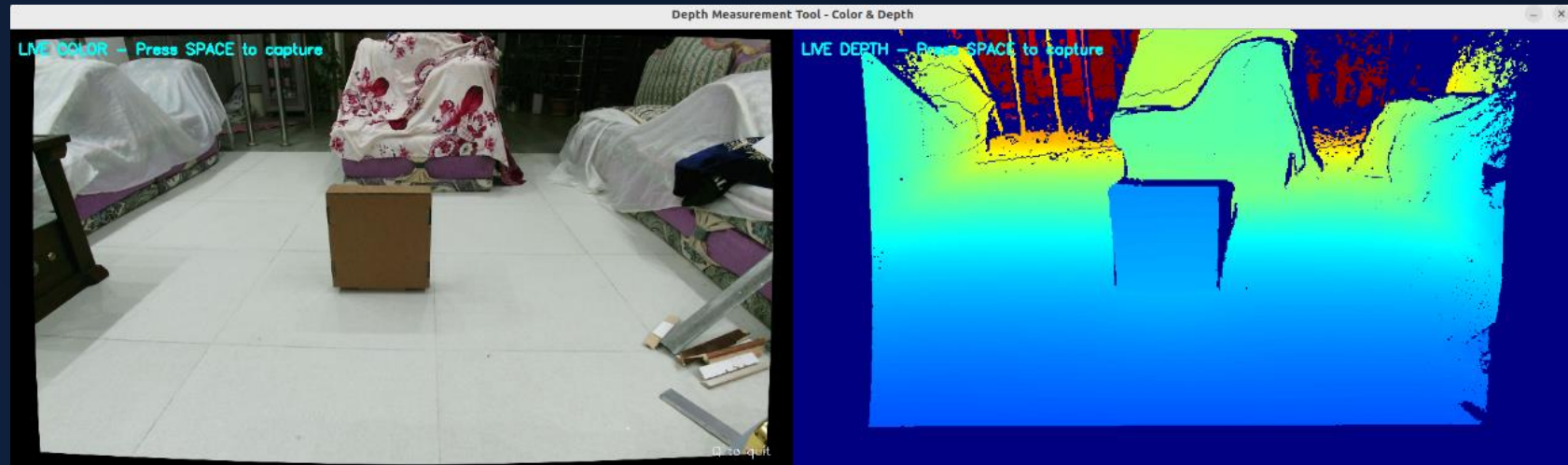
Record: Capture chessboard frames across multiple orientations using the record mode.

Calibrate: Run the offline solver to generate .yaml files for Color, IR, and Pose.



Stage 3: EXTRINSIC AND DEPTH ANALYSIS:

*Visualization of
RGB-D
alignment
after successful
extrinsic
calibration*



*Visualization of
RGB-D
alignment
after successful
extrinsic
calibration*



Stage 3: inspect the registered depth topic and PointCloud in RViz:



depth topic and Point Cloud Before Calibration



depth topic and Point Cloud After Calibration

DEPTH ERROR ANALYSIS :

METRIC	VALUE	INTERPRETATION
MEAN ERROR	~0.0 mm to ±1.5 mm	Bias effectively eliminated after correction
RMS ERROR	1.5 mm – 3.5 mm	Overall average spread of depth readings
STD DEVIATION	1.5 mm – 3.0 mm	Frame-to-frame random noise level
MIN ERROR	-10 mm to -15 mm	Worst-case underestimation (single-pixel outlier)
MAX ERROR	+10 mm to +15 mm	Worst-case overestimation (single-pixel outlier)
MEDIAN (P50)	~0.0 mm to ±1.0 mm	Typical pixel: essentially unbiased
P95 (95TH PCT.)	< 5.0 mm	95% of all pixels have error below this value

